

"event-sequence" Product Specification Version 1

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This document describes the "event-sequence" product that is used to transmit earthquake sequence definitions to PDL, which in turn forwards them to Comcat. It is assumed that the reader has a working knowledge of PDL and Comcat, which are documented at:

<https://ghsc.code-pages.usgs.gov/hazdev/pdl/index.html>
<https://earthquake.usgs.gov/data/comcat/>

An earthquake sequence is defined by a region on the Earth's surface, a start time, and an end time. All earthquakes that occur within the region, between the start time and the end time, are considered to be part of the sequence.

Type

event-sequence	The product type for event-sequence products.
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Source

us	USGS Earthquake Science Center, Moffett Field, California. Note: The network code "us" refers to the USGS National Earthquake Information Center, however event-sequence products are actually produced by the USGS Earthquake Science Center.
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Properties

An event-sequence product consists entirely of a set of properties, each of which is a key/value pair. *Property values must be given as ASCII strings, even if they are numeric in nature.* For example, a latitude might be given as the string "35.276" and *not* as the number 35.276.

eventsource	The seismic network for the mainshock. Usually, but not always, it is a 2-letter code.
eventsourcecode	The identification code for the mainshock, assigned by the seismic network. Note: The eventsource and eventsourcecode together form the mainshock's event ID. For example, if the event ID is "us7000qd1y" then the eventsource is "us" and the eventsourcecode is "7000qd1y".

review-status	The value is "reviewed" if the product was submitted manually by an analyst. This property is omitted if the product was submitted by the automatic system.
sequence-type	The type of sequence that is described by this event-sequence product. It is either "sequence" or "swarm". So far, all event-sequence products have used the value "sequence". The meaning and effect of assigning the value "swarm" has not yet been defined.
region-type	The type of region that defines this sequence. It is either "circle" or "rectangle". So far, all regions have been circular.
circle-latitude	For a circular region, the latitude of the center of the circle, in degrees.
circle-longitude	For a circular region, the longitude of the center of the circle, in degrees.
circle-radiuskm	For a circular region, the radius of the circle, in kilometers.
minimum-latitude	For a rectangular region, the latitude of the south side of the rectangle, in degrees. For a circular region, the latitude of the south side of a rectangle that contains the circle. Note: For a circular region, the associated rectangle is chosen to fit tightly about the circle (it is the <i>circumscribed</i> rectangle). The rectangle can be used to help test if an earthquake lies within the circle. If an earthquake lies outside the rectangle, then it is outside the circle. If an earthquake lies inside the rectangle, then it may or may not lie inside the circle.
maximum-latitude	For a rectangular region, the latitude of the north side of the rectangle, in degrees. For a circular region, the latitude of the north side of a rectangle that contains the circle. Note: The value of maximum-latitude is always greater than the value of minimum-latitude.
minimum-longitude	For a rectangular region, the longitude of the west side of the rectangle, in degrees. For a circular region, the longitude of the west side of a rectangle that contains the circle.

maximum-longitude	For a rectangular region, the longitude of the east side of the rectangle, in degrees. For a circular region, the longitude of the east side of a rectangle that contains the circle. Note: The value of maximum-longitude is always greater than the value of minimum-longitude. If the rectangle crosses the date line, then maximum-longitude is greater than 180. If the rectangle contains the north or south pole, then minimum-longitude is -180 and maximum-longitude is 180.
starttime	The start time of the sequence, given in ISO-8601 format. For example, "2025-06-16T20:37:41.667Z".
endtime	The end time of the sequence, given in ISO-8601 format.
eventtime	The time of the mainshock, given in ISO-8601 format. This property is optional. If included, it can be used to distinguish between foreshocks (earthquakes in the sequence that occur before the mainshock) and aftershocks (earthquakes in the sequence that occur after the mainshock).
title	A string that assigns a name to this sequence. Currently, it is the title of the mainshock with the word "Sequence" appended to the end. For example, "M 7.3 - 2025 Sand Point, Alaska Earthquake Sequence". Note: This string can contain non-ASCII characters. The character encoding is UTF-8. (The UTF-8 character encoding is defined by https://datatracker.ietf.org/doc/html/rfc3629.)
generated-by	A string that identifies the originator of the event-sequence product. For example, "USGS Aftershock Forecasting System 1.05.1848". This property is optional.

Example of an event-sequence product as it appears in Comcat

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▼ event-sequence:
  ▼ 0:
    indexid: 27898288
    indexTime: 1773521513658
    id: "urn:usgs-product:us:event-sequence:us6000rsy1:1773521511955"
    type: "event-sequence"
    code: "us6000rsy1"
    source: "us"
    updateTime: 1773521511955
    status: "UPDATE"
    ▼ properties:
      circle-latitude: "60.4966809"
      circle-longitude: "-139.7843128"
      circle-radiuskm: "60.0"
      endTime: "2027-03-14T20:41:49.288Z"
      eventsource: "us"
      eventsourcecode: "6000rsy1"
      eventtime: "2025-12-06T20:41:49.288Z"
      generated-by: "USGS Aftershock Forecasting System 1.05.1848"
      maximum-latitude: "61.0366377"
      maximum-longitude: "-138.6878451"
      minimum-latitude: "59.9567241"
      minimum-longitude: "-140.8807804"
      region-type: "circle"
      sequence-type: "sequence"
      starttime: "2025-11-06T20:41:49.288Z"
      title: "M 7.0 - 2025 Hubbard Glacier Earthquake Sequence"
      preferredWeight: 8
      contents: []
```